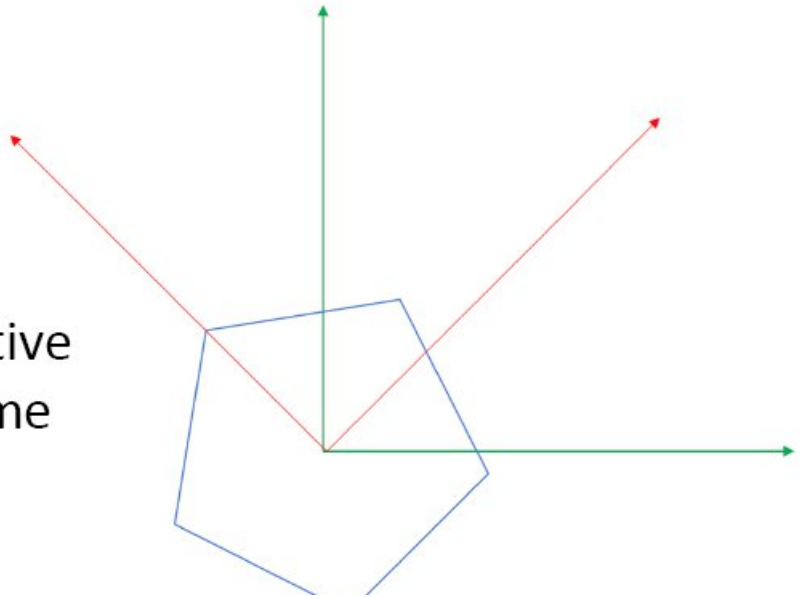
ON ROTATION & QUATERNIONS

- Basic Definitions
- Euler Angles
- Angle – Axis
- Complex numbers & Rotation
- Quaternion
- Rotation Based Quaternion representation
- Examples

ROTATION & ORIENTATION

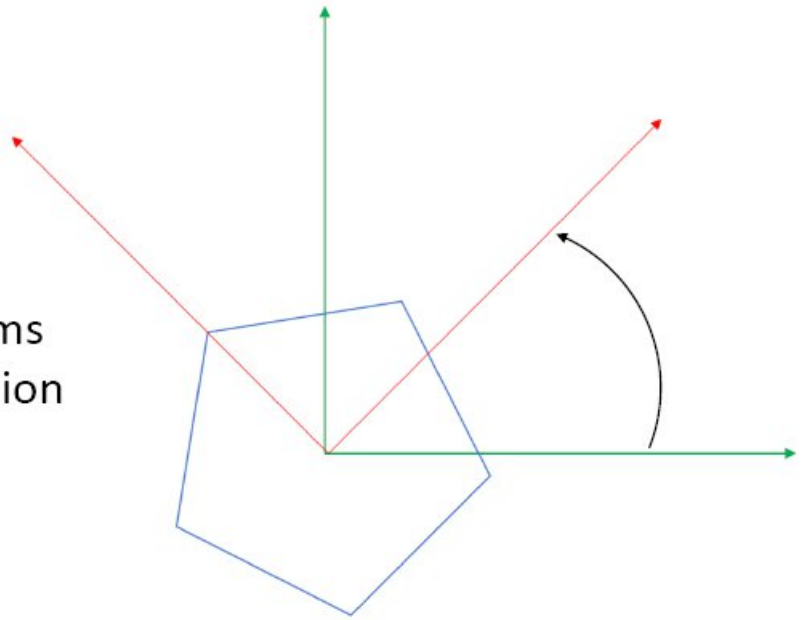
Orientation relative to reference frame



- **Orientation** of an object is described relative to some reference frame

ROTATION

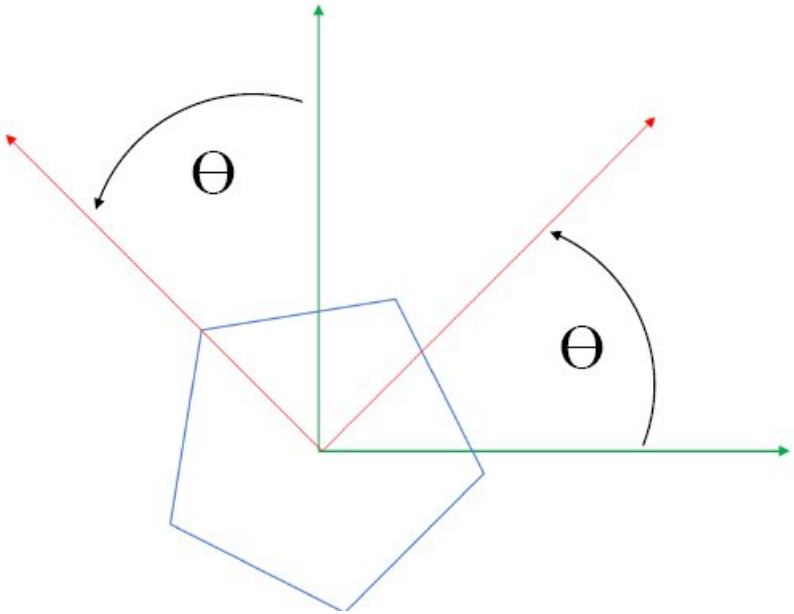
Rotation transforms from one orientation to another



- A **rotation** changes an object from one **orientation** to another

2D CASE

Rotation Angle (2D)

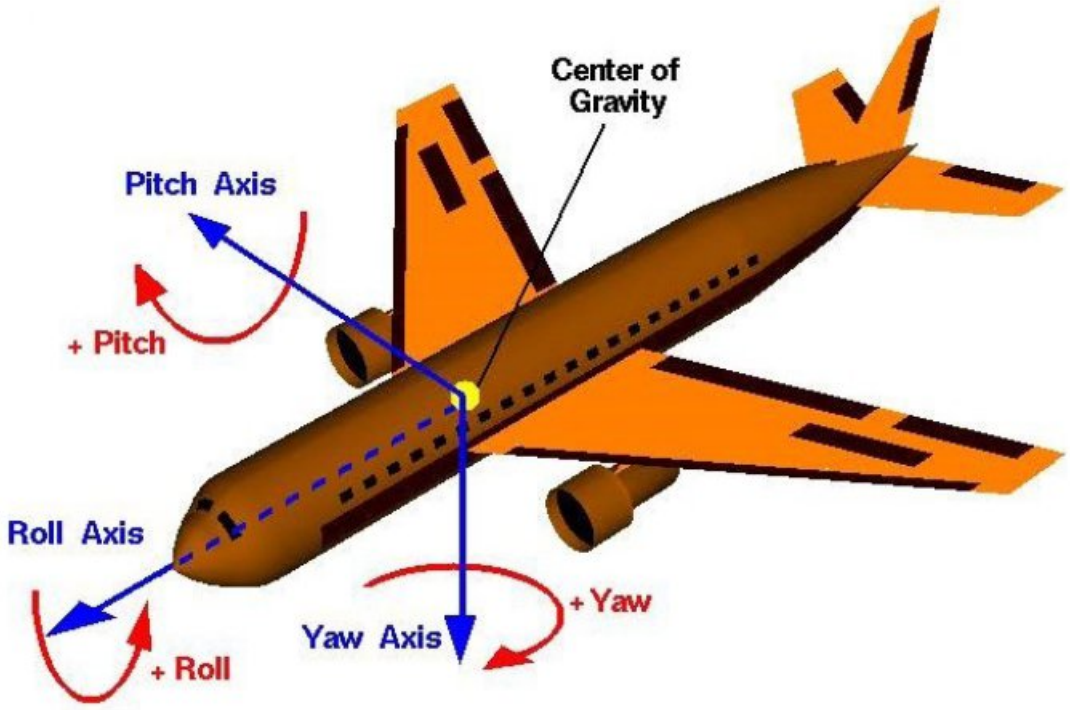


2D CASE

- Transformation
  - $(1, 0) \rightarrow (\cos \Theta, \sin \Theta)$
  - $(0, 1) \rightarrow (-\sin \Theta, \cos \Theta)$
- Rotation matrix
  - $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$

- Rotation Matrix attributes:
  - Orthogonal
    - Basis vectors are unit length
    - Dot products of basis vectors are zero
    - $M^{-1} = M^T$
    - $M^T M = \mathbf{I}$
  - Determinant is 1

3D ROTATION : Euler Angles



ZYX Euler Angles (roll, pitch, yaw)

To get from *A* to *B*:

1. Rotate ϕ about *z* axis
2. *Then* rotate θ about y axis
3. *Then* rotate ψ about x axis

$$R_z(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$R_x(\psi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & -\sin\psi \\ 0 & \sin\psi & \cos\psi \end{pmatrix}$$

$$R_{zyx}(\phi,\theta,\psi) = R_z(\phi)R_y(\theta)R_x(\psi)$$

$$R_{zyz}(\phi,\theta,\psi) = \begin{pmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & -\sin\psi \\ 0 & \sin\psi & \cos\psi \end{pmatrix}$$

Gimbal Lock : Singularity of Euler Angles

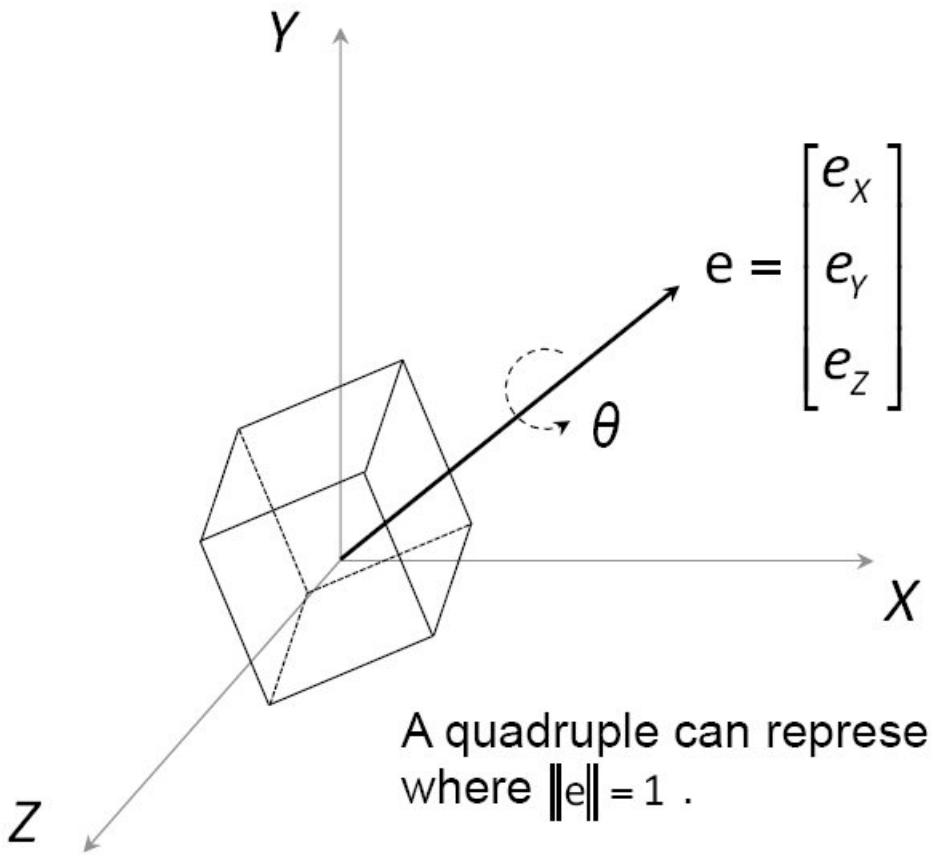
$$R_{xyz}(\alpha,\beta,\gamma) = R_x(\alpha)R_y(\beta)R_z(\gamma) =$$

$$\begin{bmatrix} \cos\beta\cos\gamma & -\cos\beta\sin\gamma & \sin\beta \\ \cos\alpha\sin\gamma+\cos\gamma\sin\alpha\sin\beta & \cos\alpha\cos\gamma-\sin\alpha\sin\beta\sin\gamma & -\cos\beta\sin\alpha \\ \sin\alpha\sin\gamma-\cos\alpha\cos\gamma\sin\beta & \cos\gamma\sin\alpha+\cos\alpha\sin\beta\sin\gamma & \cos\alpha\cos\beta \end{bmatrix}$$

Suppose we set $\beta = \pi/2$. The resulting rotation matrix is:

$$R_{xyz}(\alpha,\pi/2,\gamma) = \begin{bmatrix} 0 & 0 & 1 \\ \sin(\alpha+\gamma) & \cos(\alpha+\gamma) & 0 \\ -\cos(\alpha+\gamma) & \sin(\alpha+\gamma) & 0 \end{bmatrix}$$

Axis Angle Representation



A quadruple can represent 3D rotation, i.e., (θ, e), where $\|e\| = 1$.

Euler's theorem

Any displacement of a rigid body such that a point on the rigid body remains fixed, is equivalent to a single rotation about some axis that runs through the fixed point.

AXIS ANGLE

- Needed in various situations (e.g. camera manipulation)

about X-axis
$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

about Y-axis
$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

about Z-axis
$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

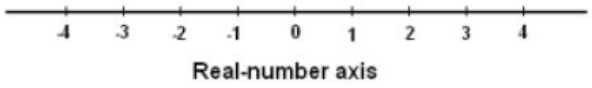
about arbitrary axis
$$R = \begin{bmatrix} \cos\theta + u_x^2(1-\cos\theta) & u_x u_y(1-\cos\theta) - u_z\sin\theta & u_x u_z(1-\cos\theta) + u_y\sin\theta \\ u_y u_x(1-\cos\theta) + u_z\sin\theta & \cos\theta + u_y^2(1-\cos\theta) & u_y u_z(1-\cos\theta) - u_x\sin\theta \\ u_z u_x(1-\cos\theta) - u_y\sin\theta & u_z u_y(1-\cos\theta) + u_x\sin\theta & \cos\theta + u_z^2(1-\cos\theta) \end{bmatrix}$$

(u_x, u_y, u_z) : axis vector

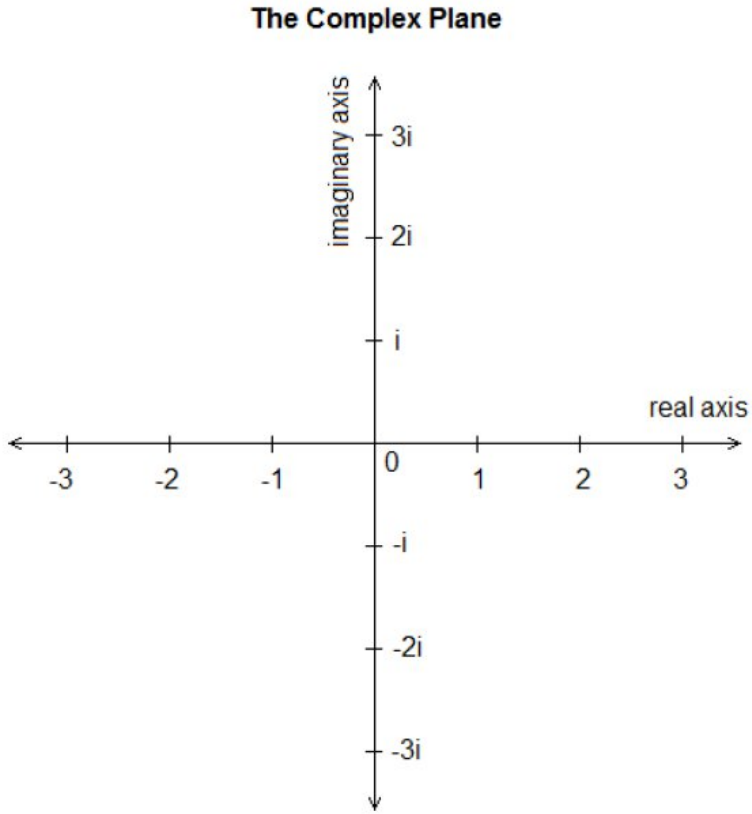
Complex Numbers & Rotation

2D numbers: $a + bi$

Where $i = \sqrt{-1}$







Complex Numbers Multiplication

Complex Numbers

- Multiply general complex number by unit complex number:

- $(x + yi)(cos\theta + isin\theta) = (xcos\theta - ysin\theta) + i(xsin\theta + ycos\theta)$

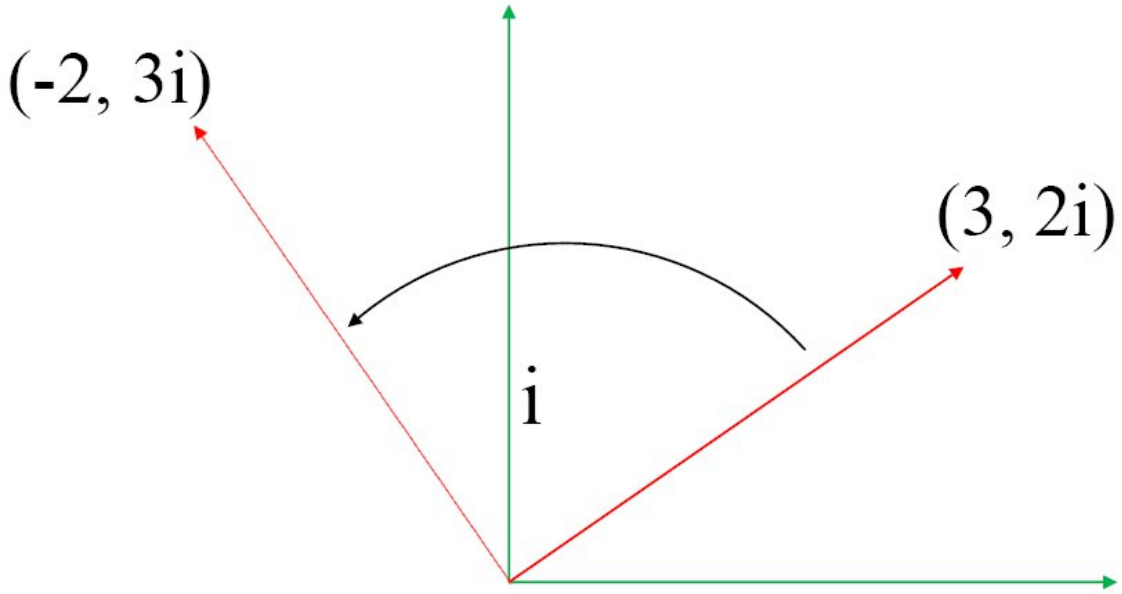
general

unit

2D rotation!

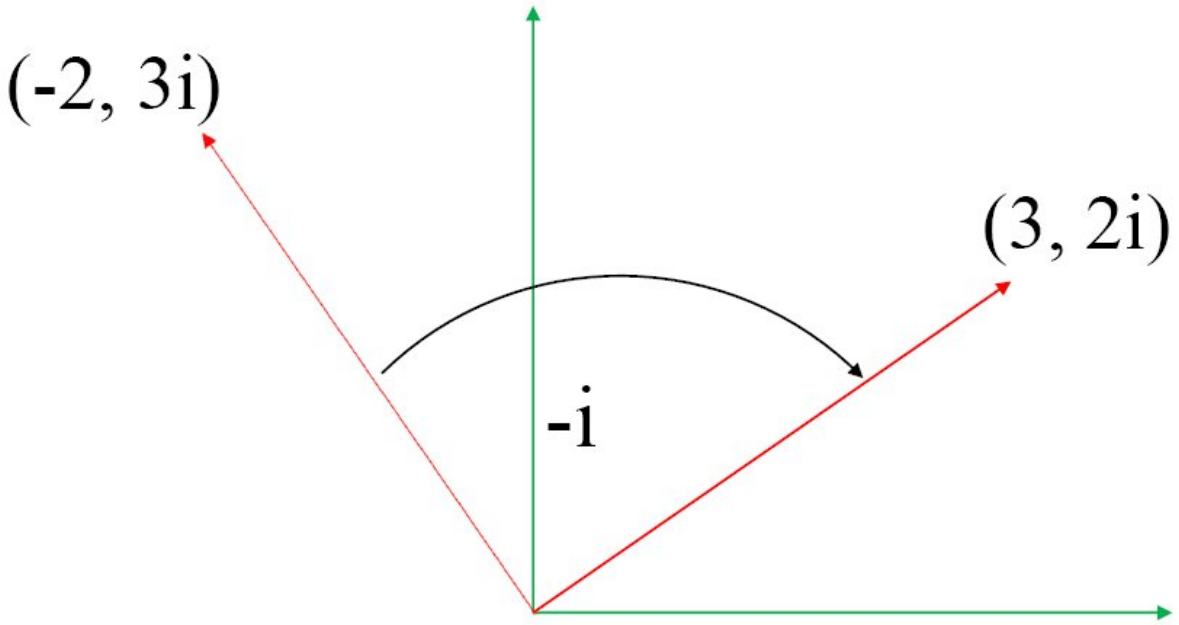
Complex Numbers Multiplication

Unit Complex Numbers are Rotations!



Complex Numbers Multiplication

Inverse Rotation



Complex Numbers Multiplication

Computational Efficiency?

- Rotation matrix vs. complex number multiplication

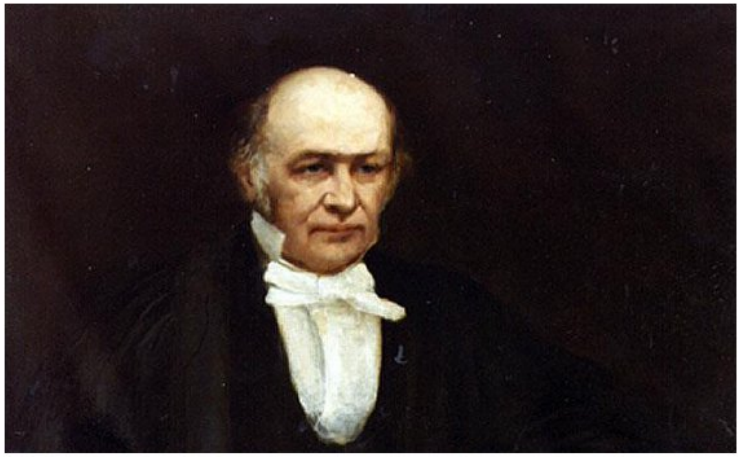
$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$

$$(p_x, p_y)(u_x, u_y)$$

3D Complex Numbers

3D Complex Numbers?

$$(a_0 + b_0 i + c_0 j)$$



3D Complex Numbers : Quaternions

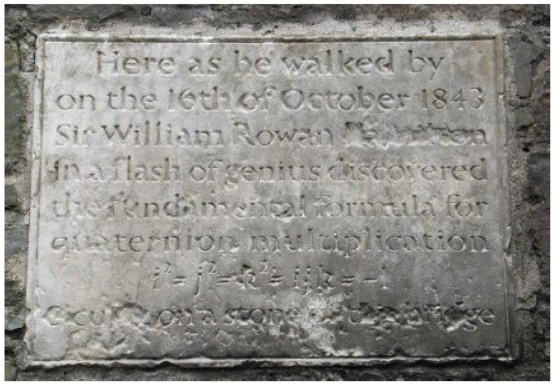
Quaternions

- Hamilton's insight: multiplication possible with 3 imaginary values

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = k \qquad jk = i \qquad ki = j$$

$$ji = -k \qquad kj = -i \qquad ik = -j$$



Quaternions extend Complex Numbers

- Suppose y = z = 0 then,

$$a + bi \rightarrow w + xi + 0j + 0k = w + xi$$

Quaternions

Generalization of complex numbers: $Q = q_0 + iq_1 + jq_2 + kq_3$

$Q = (q_0, q)$

Essentially a 4-dimensional quantity

Properties of complex dimensions:

$ii = jj = kk = ijk = -1$

$ij = -ji = k$

$jk = -kj = i$

$ki = -ik = j$

Multiplication: $QP = (q_0 + iq_1 + jq_2 + kq_3)(p_0 + ip_1 + jp_2 + kp_3)$

$QP = (p_0 q_0 - p \cdot q, p_0 q + q_0 p + p \times q)$

Complex conjugate: $Q^* = (q_0, q)^* = (q_0, -q)$

Quaternions

Let's consider the set of unit quaternions:

$$Q^2 = q_0^{\,2} + q_1^{\,2} + q_2^{\,2} + q_3^{\,2} = 1$$

This is a four-dimensional hypersphere, *i.e.* the 3-sphere S^3

The identity quaternion is: $Q = (1,0)$

Since: $$QQ^* = (q_0, q)(q_0, -q) = (q_0 q_0 - q^2, q_0 q - q_0 q + q \times q) = (1,0)$$

Therefore, the inverse of a unit quaternion is: $Q^* = Q^{-1}$

ROTATION USING QUATERNIONS

Quaternion Rotation

- Have vector **p**, unit quaternion *q*
- Treat **p** as quaternion *p* = (0, **p**)
- 3D rotation of *p* by *q* is

$$\mathbf{p'} = \mathbf{q}\,\mathbf{p}\,\mathbf{q}^{-1}$$

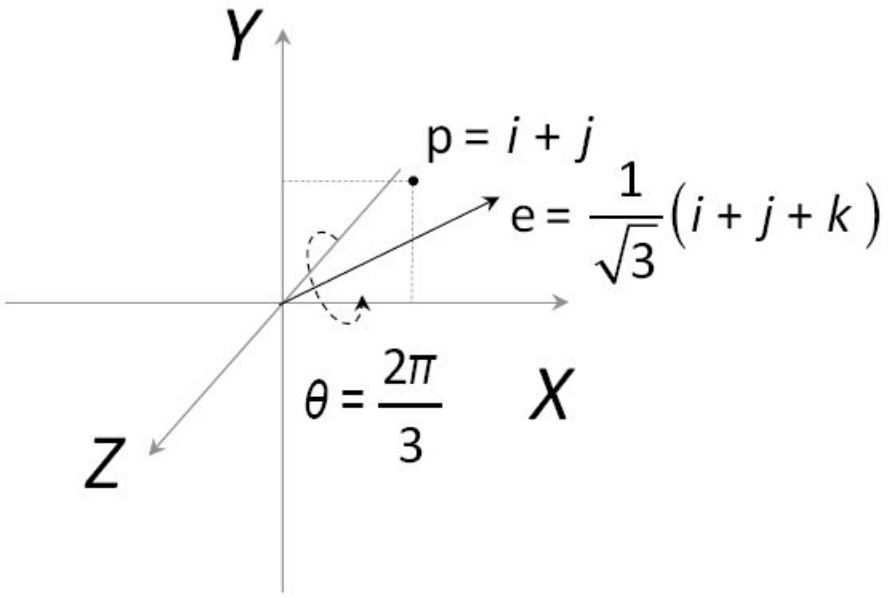
$$q = (w, v) = (\cos(\theta/2), \sin(\theta/2)n)$$

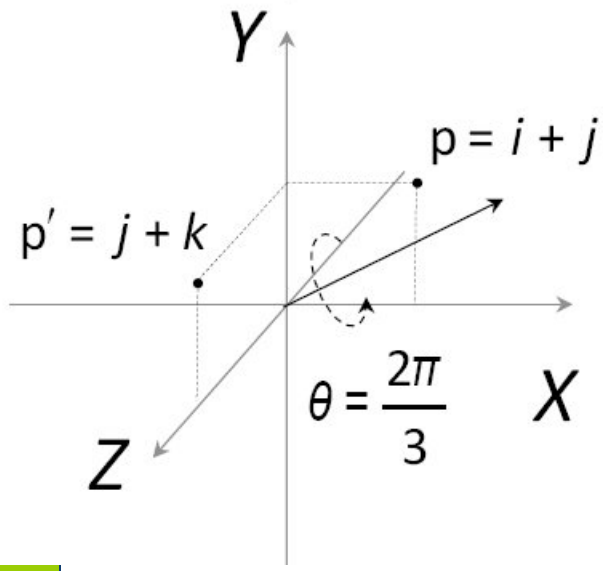
$$q^{-1} = (w, v)^{-1} = (\cos(-\theta/2), \sin(-\theta/2)n)$$

$$(w, v)^{-1} = (\cos(\theta/2), -\sin(\theta/2)n)$$

$$(w, v)^{-1} = (w, -v)$$

Example





$$X \to Y \quad Y \to Z \quad Z \to X$$

$$q = \frac{1 + i + j + k}{2} \qquad p = i + j$$

$$p' = qpq^{-1}$$

$$= \left(\frac{1 + i + j + k}{2}\right)(i + j)\left(\frac{1 - i - j - k}{2}\right)$$

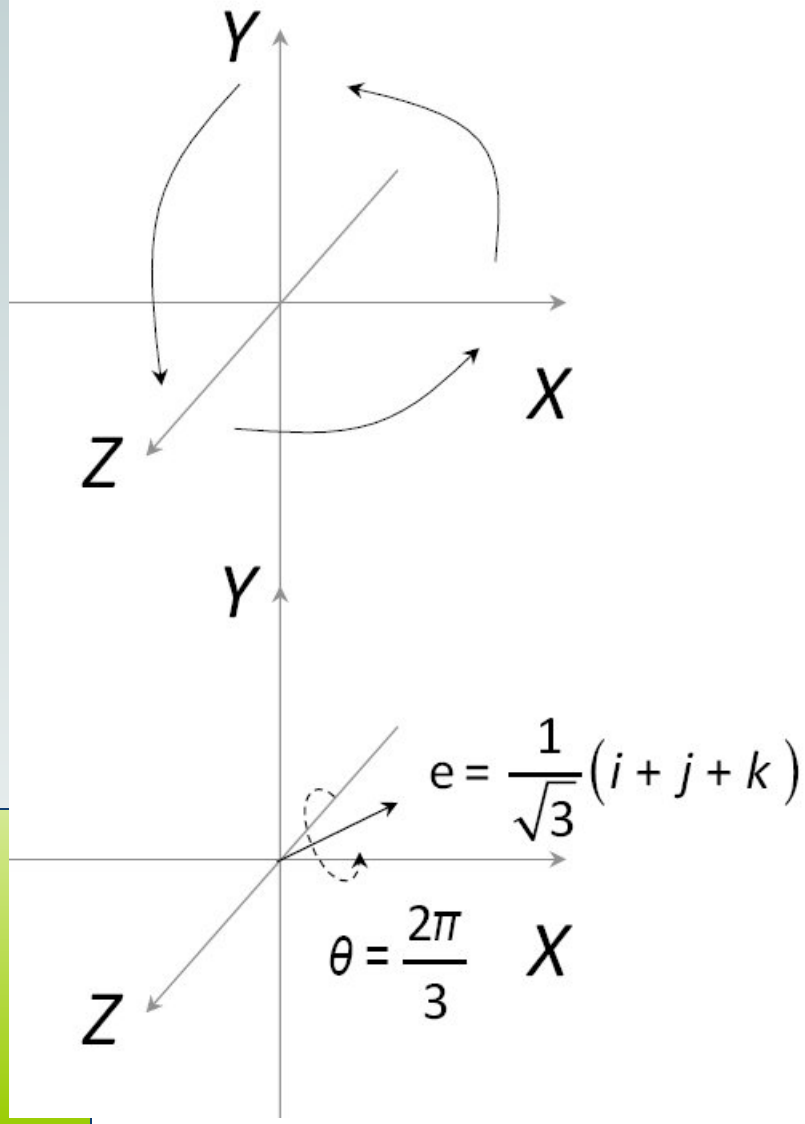
$$= \frac{1}{4}(i + j - 1 + k - k - 1 + j - i)(1 - i - j - k)$$

$$= \frac{1}{4}(2j - 2)(1 - i - j - k)$$

$$= \frac{1}{4}(2j + 2k + 2 - 2i - 2 + 2i + 2j + 2k)$$

$$= \frac{1}{4}(4j + 4k) = j + k$$

Quaternion Composition Example



$$q_1 = \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2} \qquad q_2 = \frac{\sqrt{2}}{2} + k\frac{\sqrt{2}}{2}$$

$$q_{12} = q_1 q_2$$

$$= \left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right)$$

$$= \frac{1}{2} + \frac{1}{2}i + \frac{1}{2}j + \frac{1}{2}k$$

$$= \frac{1}{2} + \frac{1}{2}i + \frac{1}{2}j + \frac{1}{2}k$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}\left(\frac{1}{\sqrt{3}}i + \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k\right)$$

$$\cos\frac{\theta}{2} = \frac{1}{2} \quad \sin\frac{\theta}{2} = \frac{\sqrt{3}}{2} \longrightarrow \theta = \frac{2\pi}{3}$$

Rotating about (1,1,1) axis with 120 degrees.

Quaternions Advantages

*__Avoid gimbal lock__: Euler angles can fail when two rotation axes align, a situation known as gimbal lock.

*__Improve computational efficiency__: Quaternions require fewer operations than rotation matrices,

*__Reduce memory usage__: You only need four values to represent a quaternion, compared to nine for a rotation matrix.

*__Enable smooth interpolation__: smoothly transition between orientations.

*__Numerical stability__: Quaternions are less prone to rounding errors than traditional rotation methods during repeated transformations.